

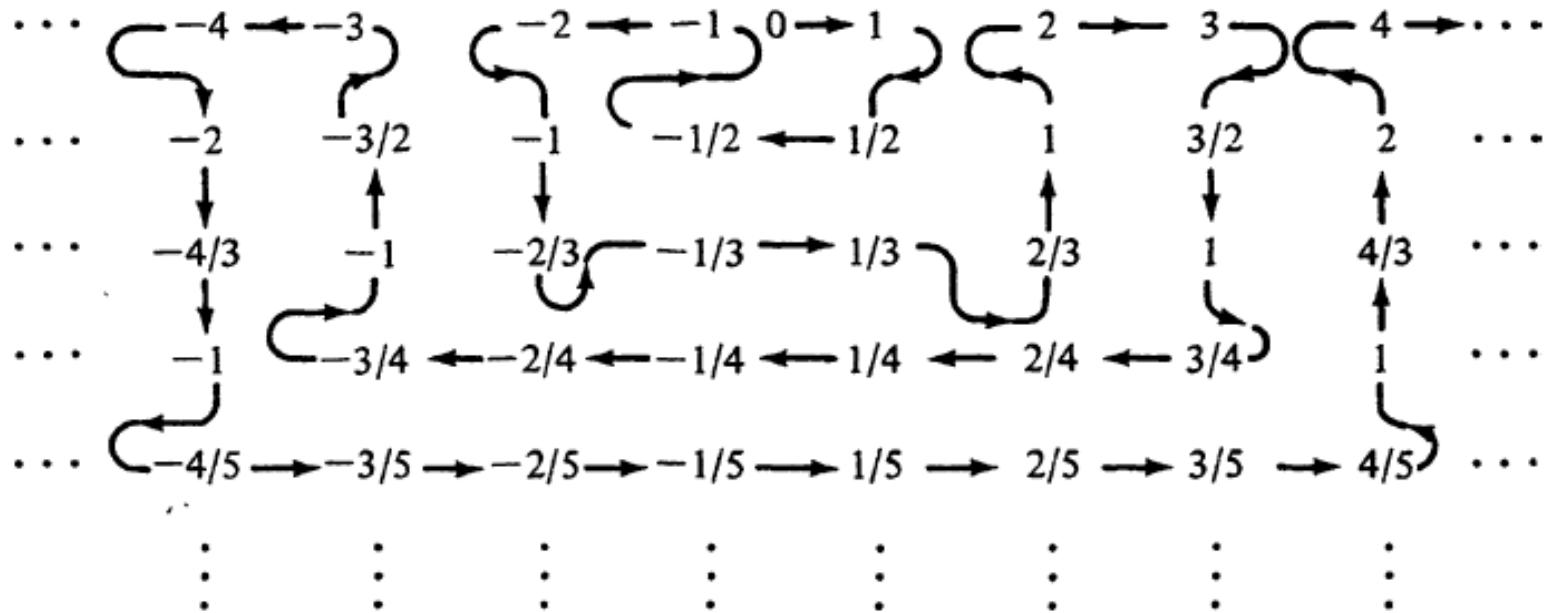
Math 421

Monday, October 20

Announcements

- Grace's Drop-In Hours
 - Wednesday 12-1 pm VV 311
 - Thursday 9-11 am VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table 4
- Reading Quiz on $\liminf/\limsup$ due Wednesday
- Homework 6 Due Friday

Example – Sequence of rational numbers



Activity

Suppose (s_n) is a sequence of positive numbers (i.e $s_n > 0$) such that $\inf\{s_n : n \in \mathbb{N}\} = 0$. Prove that no matter how (s_n) itself behaves, it has a subsequence converging to 0.

- ❖ Use Theorem 11.2 Part 1 - If t is in $\mathbb{R}$, then there is a subsequence of (s_n) converging to t if and only if the set $\{n \in \mathbb{N} : |s_n - t| < \epsilon\}$ is infinite for all $\epsilon > 0$.
- ❖ Use a proof by contradiction (what are your new hypotheses?)
 - Case 1: The set is non-empty (Hint: if a set is non-empty and finite, the infimum is equal to the minimum of the set).
 - Case 2: The set is empty.

Theorems involving subsequences

Theorem 11.3: If the sequence (s_n) converges, then every subsequence converges to the same limit.

Theorem 11.4: Every sequence (s_n) has a monotonic subsequence.

Discussion Question: Convince yourselves these statements are true. It could be helpful to play around with some specific examples.

Theorem 11.3 – Discussion

Theorem 11.3: If the sequence (s_n) converges, then every subsequence converges to the same limit.

Discussion.

Theorem 11.4 – Discussion

Theorem 11.4: Every sequence (s_n) has a monotonic subsequence.

Discussion.

Bolzano-Weierstrass Theorem

Thm: Every bounded sequence has a convergent subsequence.

Discussion Question: Why is this theorem true?

Subsequential Limit

Def. Let (s_n) be a sequence in $\mathbb{R}$. A **subsequential limit** is any real number or symbol $+\infty$ or $-\infty$ that is the limit of some subsequence of (s_n) .

Example.

Activity

For each of the following sequences, determine its set of subsequential limits.

1. (s_n) where $s_n = \frac{1}{n}$.
2. (a_n) where $a_n = n^2(-1)^n$.
3. (b_n) where $b_n = \sin\left(\frac{n\pi}{3}\right)$.
4. (r_n) a list of all rational numbers.
5. (c_n) where $c_n = n(1 + (-1)^n)$.

Activity

Consider the sequence defined by $s_n = \frac{1}{n}$.

1. Define the sets $S_N = \{s_n : n > N\}$, the “tails” of the sequence. Write out a couple of these sets, $N = 1, 5, 10$ for example.
2. What is the relationship between these sets?
3. Define $u_N = \inf\{s_n : n > N\}$ and $v_N = \sup\{s_n : n > N\}$. Find these values for the sets you looked at in 1.
4. What do you notice about the relationship between the u_N and the v_N ?
5. What is $u = \lim_{N \rightarrow \infty} u_N$? $v = \lim_{N \rightarrow \infty} v_N$?

The sets $\{s_n : n > N\} = S_N$

lim sup **and** lim inf

Def. Let (s_n) be a sequence in $\mathbb{R}$. We define

$$\limsup s_n = u = \lim_{N \rightarrow \infty} u_N = \lim_{N \rightarrow \infty} \sup\{s_n : n > N\}$$

and

$$\liminf s_n = v = \lim_{N \rightarrow \infty} v_N = \lim_{N \rightarrow \infty} \inf\{s_n : n > N\}$$

Example. Let $s_n = \frac{1}{n}$.

$$\lim s_n = \liminf s_n = \limsup s_n$$

Thm: Let (s_n) be a sequence in $\mathbb{R}$.

1. If $\lim s_n$ is defined (as a real number or $\pm\infty$) then

$$\liminf s_n = \lim s_n = \limsup s_n .$$

2. If $\liminf s_n = \limsup s_n$, then $\lim s_n$ is defined and

$$\lim s_n = \liminf s_n = \limsup s_n .$$

Theorem 11.8

Thm. Let (s_n) be any sequence in $\mathbb{R}$, and let S denote the set of subsequential limits of (s_n) .

1. S is nonempty.
2. $\sup S = \limsup s_n$ and $\inf S = \liminf s_n$.
3. $\lim s_n$ exists if and only if S has exactly one element, namely $\lim s_n$.

❖ Convince yourselves this theorem is true.

Discussion Question

Convince yourselves the following statements are true, provide examples where appropriate:

- ❖ If $L = \limsup s_n \neq \infty$, then for every $\alpha > L$, the set $\{n : s_n > \alpha\}$ is finite.
- ❖ The set $\{n : s_n > \limsup s_n\}$ can be infinite.
- ❖ If $\liminf s_n < \limsup s_n$, the set
$$\{n : \liminf s_n \leq s_n \leq \limsup s_n\}$$
can be empty.